

# Secure Multi-Party Computation (SMC)

## Introduction

The focus of this task was to implement a Secure Multi-Party Computation (SMC) framework that enables financial institutions to compute their total account balance without revealing individual balances. This was achieved using Shamir's Secret Sharing (SSS) to split sensitive data into secure shares and Lagrange interpolation to reconstruct the total collaboratively. The aim was to ensure privacy, accuracy, and scalability in these computations.

This document explains the work done, improvements made, and how the implementation was tested and validated. It also highlights how the demonstration task showcases the practical use of SMC.

## Objective

The goal was to securely compute the total account balance across financial institutions while keeping their individual balances private. Specifically, the task involved:

- Using Shamir's Secret Sharing to split balances into secure shares.
- Refining the Lagrange interpolation process for reconstructing totals efficiently.
- Adding Gaussian noise to enhance privacy.
- Testing the framework under different configurations to ensure it works correctly and can scale.

## Implementation Overview

### Shamir's Secret Sharing (SSS):

Each institution's balance is split into shares using a polynomial function. The constant term in the polynomial is the balance, while the other coefficients are random.

A threshold (e.g., 3 out of 5 shares) is required to reconstruct the total. Fewer than the threshold gives no meaningful information about the balance.

### Share Generation:

Shares were generated by evaluating the polynomial at different x-values (e.g.,  $x = 1, 2, 3, \dots$ ).

Gaussian noise was added to the shares to enhance privacy. This ensures that even if some shares are exposed, the original balance cannot be reverse-engineered.

### Reconstruction Using Lagrange Interpolation:

Selected shares (equal to or greater than the threshold) were used to reconstruct the total account balance by evaluating the polynomial at  $x=0$ .

The interpolation process was refined to make it faster and better suited for larger datasets or higher thresholds.

## Enhancements and Features

### Gaussian Noise for Privacy:

Noise was added during share generation to make the framework more secure by ensuring shares alone could not reveal sensitive data.

### Refining Lagrange Interpolation:

The reconstruction process was optimized to handle more data efficiently, reducing computational overhead and improving scalability.

### Flexible Configurations:

The system supports different numbers of shares and thresholds, making it adaptable for various real-world scenarios involving collaborative financial computations.

## Testing and Validation

### Testing Scenarios:

The implementation was tested with different numbers of institutions (e.g., 3, 5) and thresholds (e.g., requiring 3 shares out of 5).

Gaussian noise was temporarily removed during testing to ensure that reconstructed totals matched actual totals, confirming the mathematical correctness of the implementation.

### Validation Results:

The reconstructed totals consistently matched the expected totals when the threshold was met.

Testing also demonstrated that the framework is reliable under different configurations and capable of scaling to handle more data or participants.

## Demonstration Task

A demonstration notebook was created to show how the SMC framework works in practice. It highlights key use cases like:

Secure Collaboration: Institutions compute their total account balance without exposing individual balances.

Privacy Preservation: The use of Shamir's Secret Sharing and Gaussian noise ensures that sensitive data remains secure.

Validation of Results: The notebook demonstrates that the reconstructed totals are accurate when the required threshold of shares is met.

## Key Learnings and Challenges

### Key Learnings:

- Privacy-preserving computations are feasible and scalable using cryptographic techniques like SSS.
- The balance between privacy and accuracy is critical, especially when integrating noise for additional security.

### Challenges:

- Ensuring the accuracy of reconstructed totals under different configurations.
- Optimizing the Lagrange interpolation process to handle large datasets efficiently.

These challenges were addressed through iterative testing and refinements.

### **Conclusion**

The SMC framework successfully demonstrated how financial institutions can compute a total account balance collaboratively while keeping individual balances private. By using Shamir's Secret Sharing, Gaussian noise, and optimized Lagrange interpolation, the implementation meets the goals of privacy, accuracy, and scalability. The demonstration task provided a practical example of the framework in action, showing how such methods can be applied to real-world scenarios.